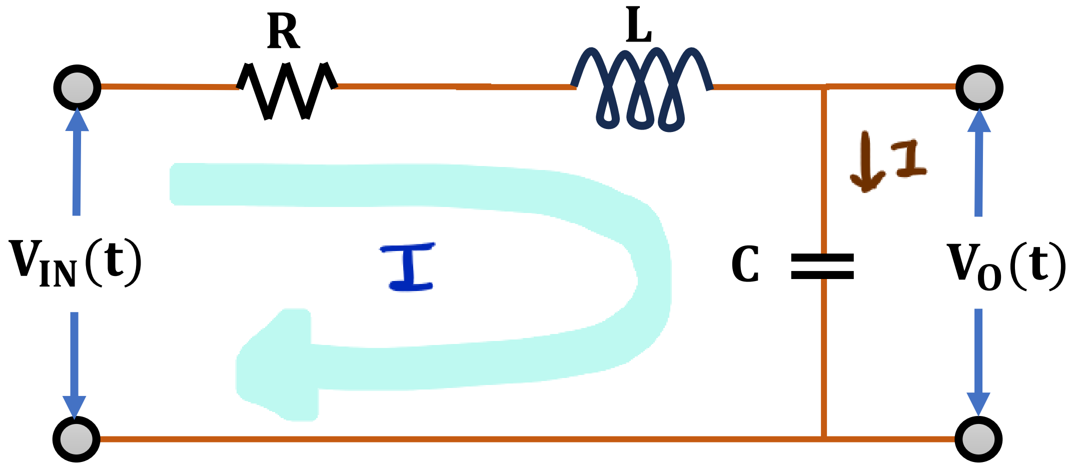
Mathematical Models

Types of Mathematical Models in Control Systems

- Differential Equation Model
- Transfer Function Model
- State Space Model

Differential Equation Model and Transfer Function Model of RLC Circuit



→ Apply KVL in loop.

$$\Rightarrow V_{in}(t) = V_R + V_L + V_o(t)$$

$$\Rightarrow V_{in}(t) = IR + L\frac{dI}{dt} + V_o(t)$$

[→ Through Capacitor $I = C\dfrac{dV_o(t)}{dt}$]

$$\Rightarrow V_{in}(t) = RC\frac{dV_o(t)}{dt} + LC\frac{d^2V_o(t)}{dt} + V_o(t)$$

$$\boxed{LC\frac{d^2V_o(t)}{dt^2} + RC\frac{dV_o(t)}{dt} + V_o(t) = V_{in}(t)} \quad \text{--- (A)}$$

⇝ eqⁿ (A) is differential eqⁿ of RLC circuit.

⇝ To get Transfer function, apply Laplace X'form to eqⁿ (A).

$$\Rightarrow LC\, s^2 V_o(s) + RC\, s V_o(s) + V_o(s) = V_{in}(s)$$

$$\Rightarrow \boxed{\frac{V_o(s)}{V_{in}(s)} = \frac{1}{LCs^2 + RCs + 1}} \quad \text{--- (B)}$$

⇝ eqⁿ (B) is transfer function of RLC circuit.

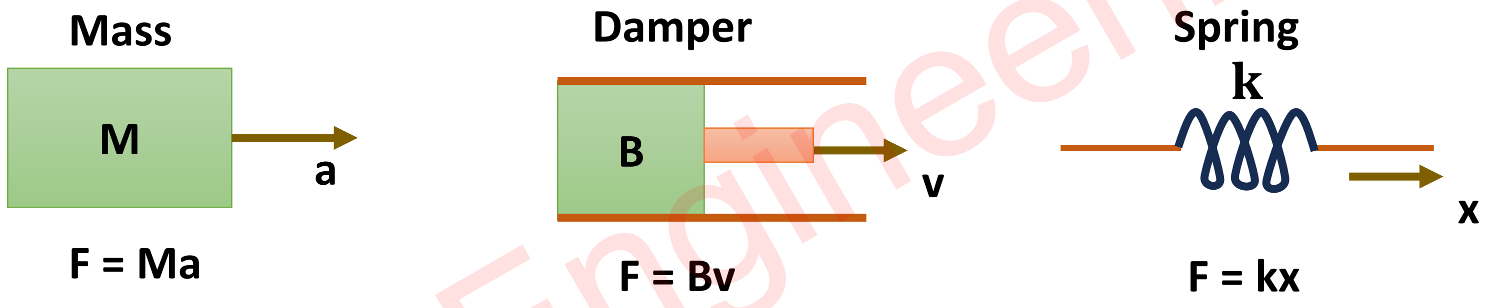
Mechanical Systems

Types of Mechanical Systems

- Based on the types of Motion, Mechanical Systems are classified into two types:
  1. Translational Mechanical System
  2. Rotational Mechanical System

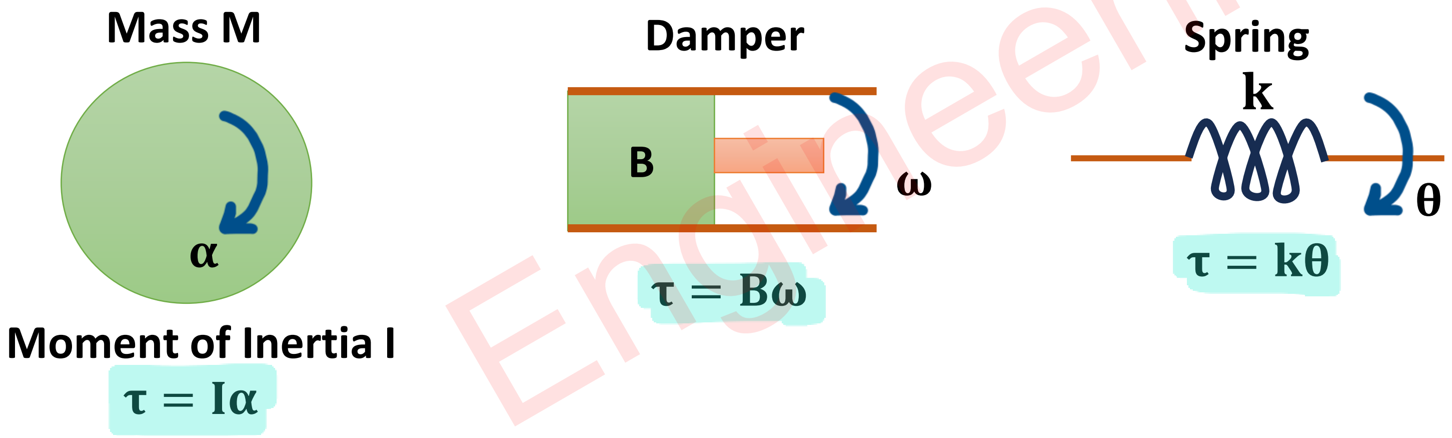
Translational Mechanical System

- In this Mechanical System, object motion happens in a straight line.
- Parameters of Translational Mechanical System:
  - Displacement x [Unit = Meter]
  - Velocity $\mathrm{v} = \frac{\mathrm{dx}}{\mathrm{dt}}$ [Unit = Meter/Sec]
  - Acceleration $\mathrm{a} = \frac{\mathrm{dv}}{\mathrm{dt}} = \frac{\mathrm{d^2x}}{\mathrm{dt^2}}$ [Unit = $\mathrm{Meter/Sec^2}$]
  - Mass M [Unit = Kg]
  - Force F [Unit = N]



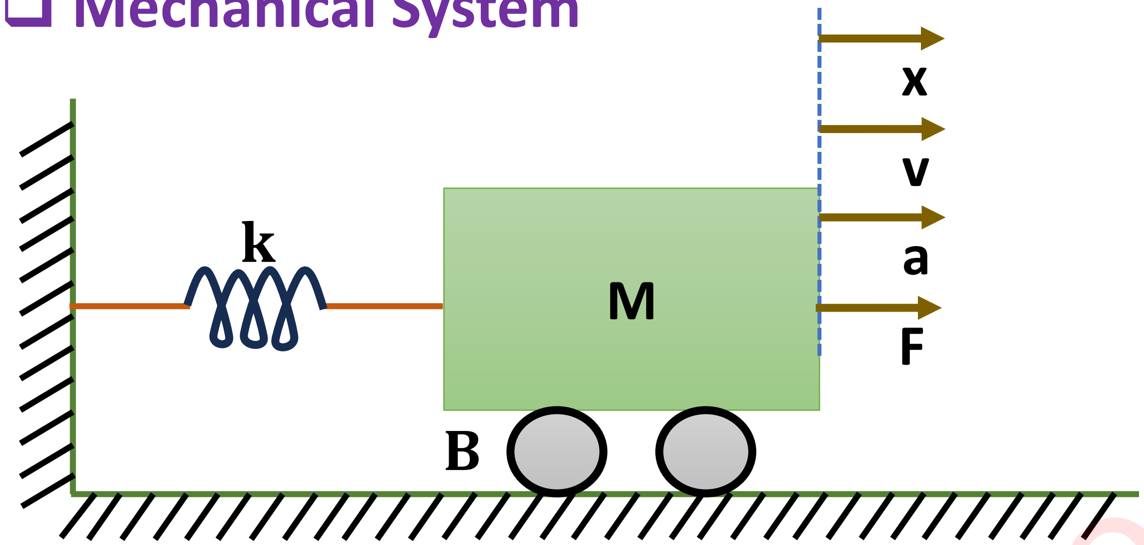
Rotational Mechanical System

- In this Mechanical System, object motion happens in a Circular Line.
- Parameters of Rotational Mechanical System:
  - Angular Displacement θ [Unit = Rad]
  - Angular Velocity $\omega = \frac{d\theta}{dt}$ [Unit = Rad/Sec]
  - Angular Acceleration $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$ [Unit = Rad/Sec^2]
  - Moment of Inertia I or J [Unit = Kg m^2]
  - Torque τ [Unit = Nm]



Force Voltage Analogy

❑ Mechanical System



- Force distribution in Spring,

$$\Rightarrow F_K = kx$$

- Force distribution in Damper,

$$\Rightarrow F_B = Bv$$

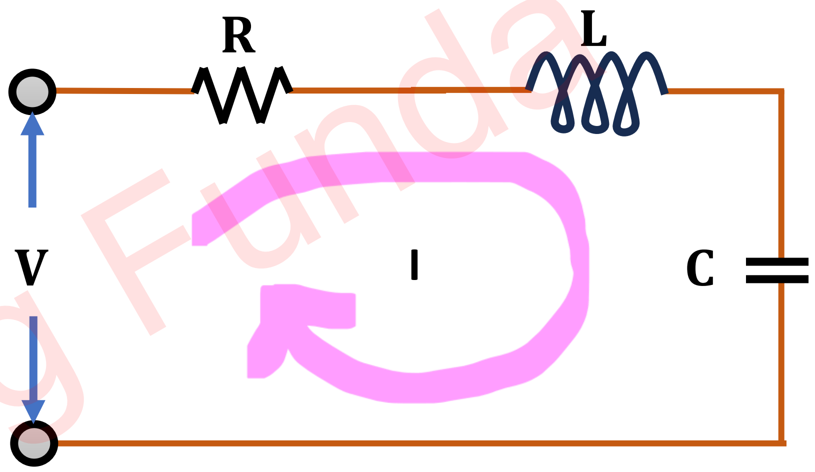
- Force distribution in Mass,

$$\Rightarrow F_M = Ma$$

- Total Force distribution,

$$\Rightarrow F = F_k + F_B + F_M = kx + Bv + Ma$$

❑ Electrical System



- Apply KVL in Loop I,

$$\Rightarrow V = V_R + V_L + V_C$$

$$\Rightarrow V = IR + L\frac{dI}{dt} + \frac{1}{C}\int Idt \qquad \left[\therefore \int Idt = Q\right]$$

$$\Rightarrow V = IR + L\frac{dI}{dt} + \frac{Q}{C}$$

❑ Mechanical System

$$\Rightarrow F = kx + Bv + Ma$$

$$\Rightarrow F = kx + B\frac{dx}{dt} + M\frac{d^2x}{dt^2}$$

- Apply Laplace Transform,

$$\Rightarrow F = kx + BSx + MS^2x$$

❑ Electrical System

$$\Rightarrow V = IR + L\frac{dI}{dt} + \frac{Q}{C} \qquad \left[\therefore I = \frac{dQ}{dt}\right]$$

$$\Rightarrow V = R\frac{dQ}{dt} + L\frac{d^2Q}{dt^2} + \frac{Q}{C}$$

- Apply Laplace Transform,

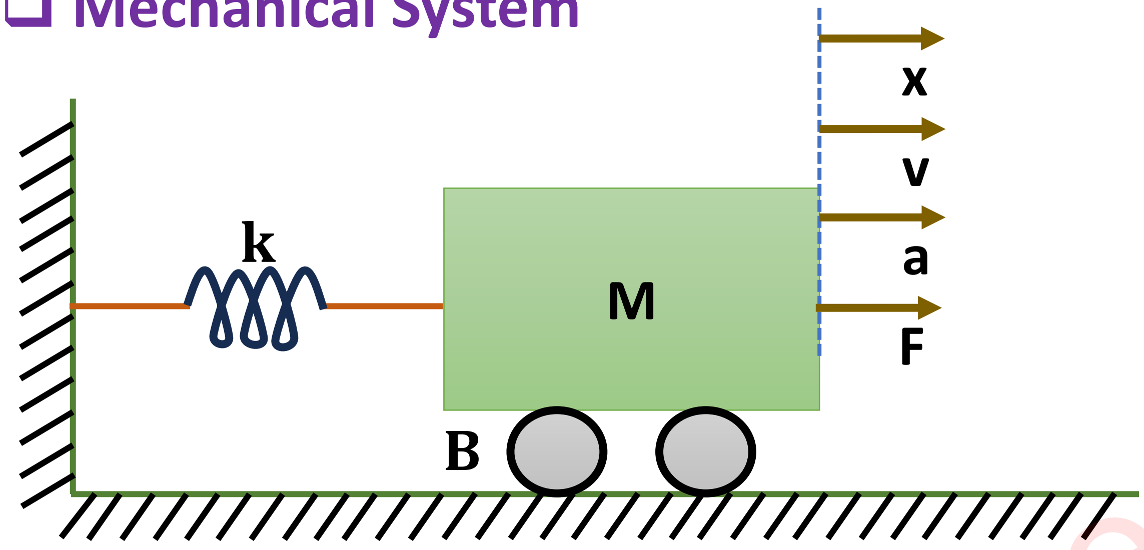
$$\Rightarrow V = RSQ + LS^2Q + \frac{Q}{C}$$

❑ Force Voltage Analogy

Mechanical System	Electrical System
❖ Force F	❖ Voltage V
❖ Mass M	❖ Inductance L
❖ Damping Constant B	❖ Resistance R
❖ Spring Constant k	❖ Reciprocal of Capacitance (1/C)
❖ Distance x	❖ Charge Q
❖ Velocity $v = \frac{dx}{dt}$	❖ Current $I = \frac{dQ}{dt}$

Force Current Analogy

❑ Mechanical System



- Force distribution in Spring,

  $\Longrightarrow F_K = kx$

- Force distribution in Damper,

  $\Longrightarrow F_B = Bv$

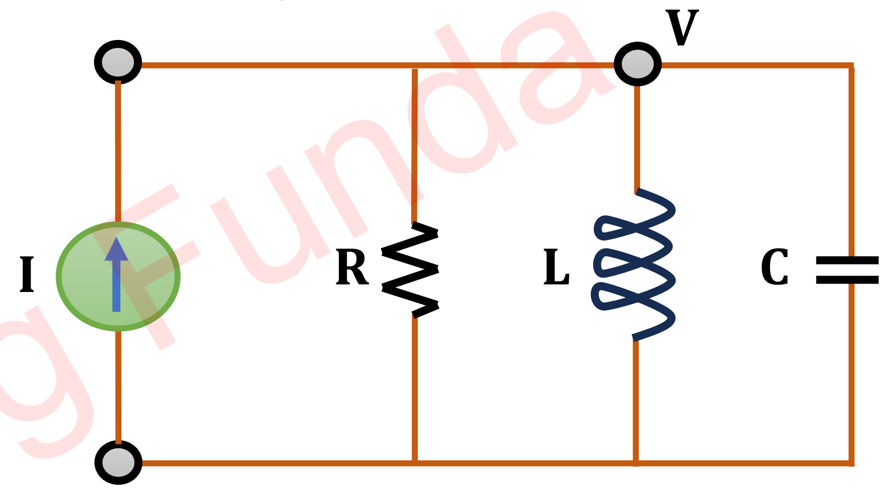
- Force distribution in Mass,

  $\Longrightarrow F_M = Ma$

- Total Force distribution,

  $\Longrightarrow F = F_k + F_B + F_M = kx + Bv + Ma$

❑ Electrical System



- Apply KCL Node V,

$$\Longrightarrow I = I_R + I_L + I_C$$

$$\Longrightarrow I = \frac{V}{R} + \frac{1}{L}\int Vdt + C\frac{dV}{dt}$$

$$\left[\therefore V = \frac{d\phi}{dt}\right] \quad \left[\therefore \phi = \int Vdt\right]$$

$$\Longrightarrow I = \frac{1}{R}\frac{d\phi}{dt} + \frac{1}{L}\phi + C\frac{d^2\phi}{dt^2}$$

Mechanical System

$$\Rightarrow F = F_k + F_B + F_M = kx + Bv + Ma$$

$$\Rightarrow F = kx + B\frac{dx}{dt} + M\frac{d^2x}{dt^2}$$

- Apply Laplace Transform,

$$\Rightarrow F = kx + BSx + MS^2x$$

Electrical System

$$\Rightarrow I = \frac{1}{R}\frac{d\phi}{dt} + \frac{1}{L}\phi + C\frac{d^2\phi}{dt^2}$$

- Apply Laplace Transform,

$$\Rightarrow I = \frac{1}{R}S\phi + \frac{1}{L}\phi + CS^2\phi$$

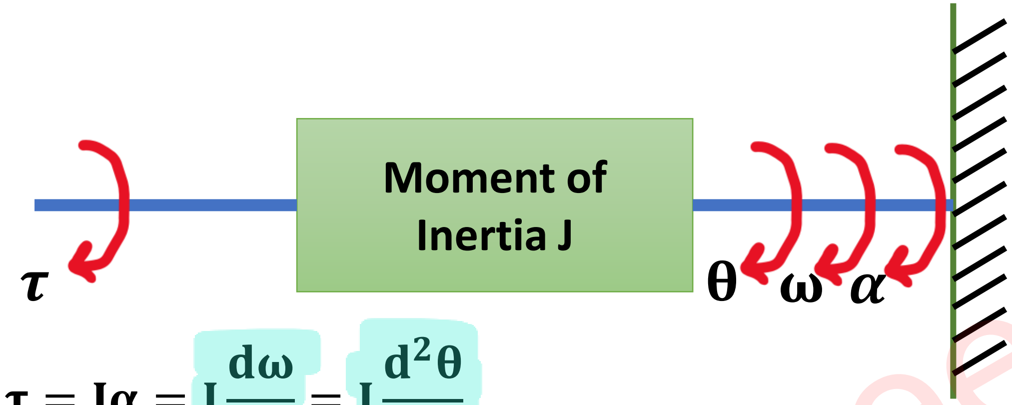
Force Current Analogy

Mechanical System	Electrical System
❖ Force F	❖ Current I
❖ Mass M	❖ Capacitance C
❖ Damping Constant B	❖ Reciprocal of Resistance (1/R)
❖ Spring Constant k	❖ Reciprocal of Inductance (1/L)
❖ Distance x	❖ Flux ϕ
❖ Velocity $v = \frac{dx}{dt}$	❖ Voltage $V = \frac{d\phi}{dt}$

Torque Voltage Analogy

❑ Mechanical System

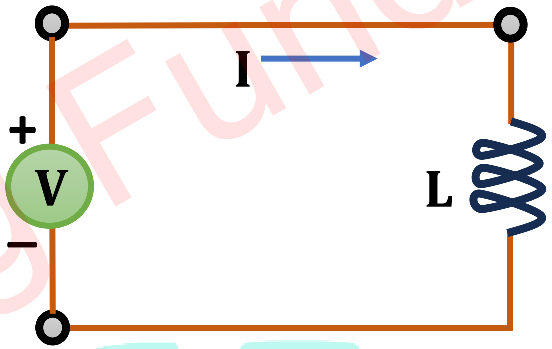
- Input = Torque τ
- Output = Angular Velocity ω



$$\Rightarrow \tau = J\alpha = J\frac{d\omega}{dt} = J\frac{d^2\theta}{dt^2}$$

❑ Electrical System

- Input = Voltage V
- Output = Current I

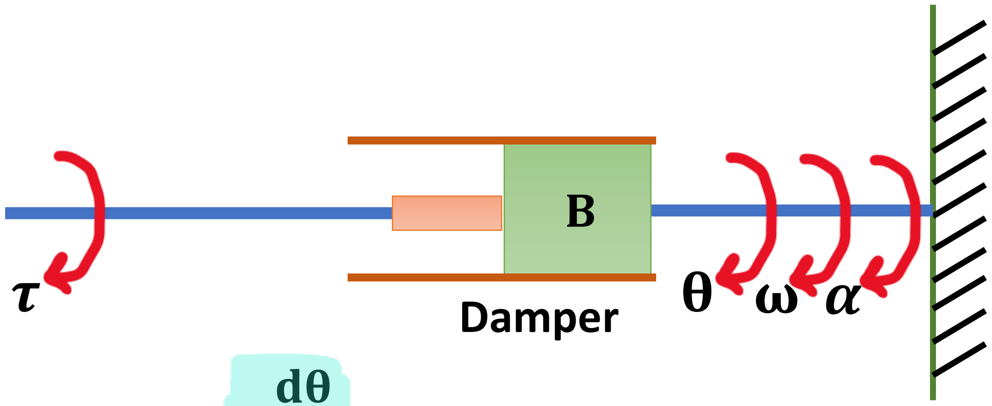


$$\Rightarrow V = L\frac{dI}{dt} = L\frac{d^2Q}{dt^2}$$

❑ Torque Voltage Analogy

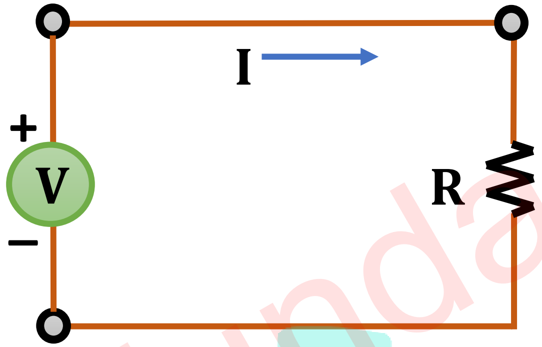
Mechanical System	Electrical System
❖ Torque τ	❖ Voltage V
❖ Angular Velocity ω	❖ Current I
❖ Moment Inertia J	❖ Inductance L
❖ Angular Displacement θ	❖ Charge Q

❑ Mechanical System



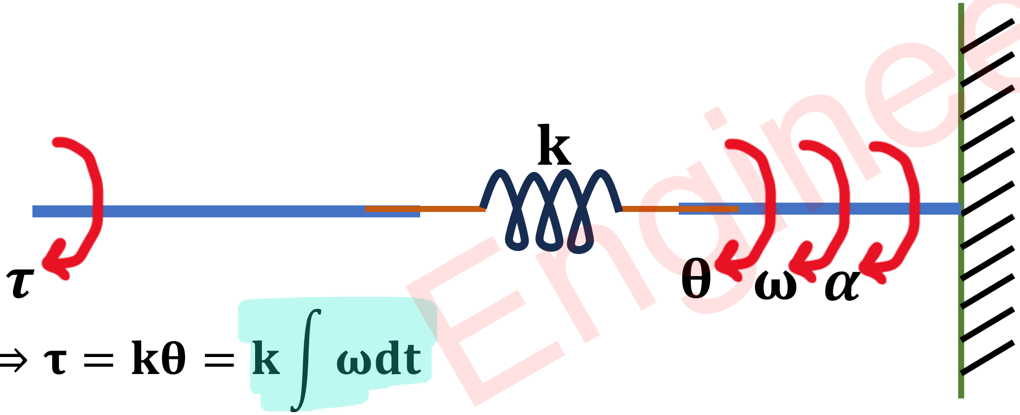
$$\Rightarrow \tau = B\omega = B\frac{d\theta}{dt}$$

❑ Electrical System

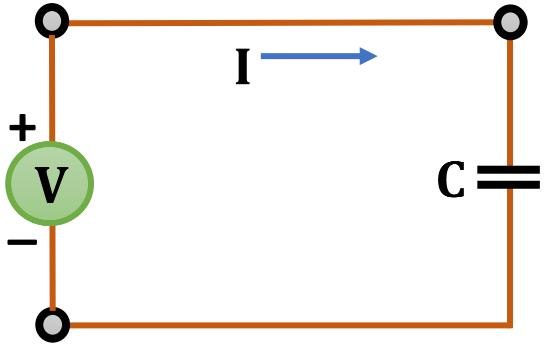


$$\Rightarrow V = IR = R\frac{dQ}{dt}$$

Mechanical System	Electrical System
❖ Damping Constant B	❖ Resistance R



$$\Rightarrow \tau = k\theta = k\int \omega dt$$



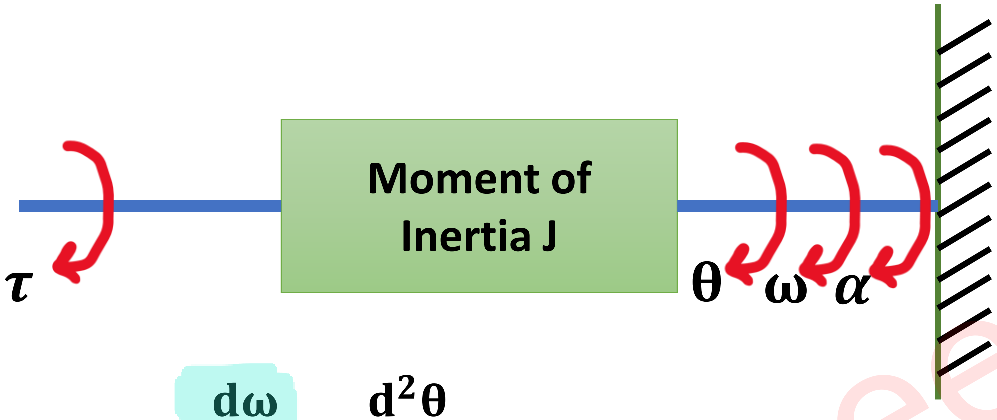
$$\Rightarrow V = \frac{Q}{C} = \frac{1}{C}\int Idt$$

Mechanical System	Electrical System
❖ Spring Constant k	❖ Reciprocal of Capacitance (1/C)

Torque Current Analogy

❑ Mechanical System

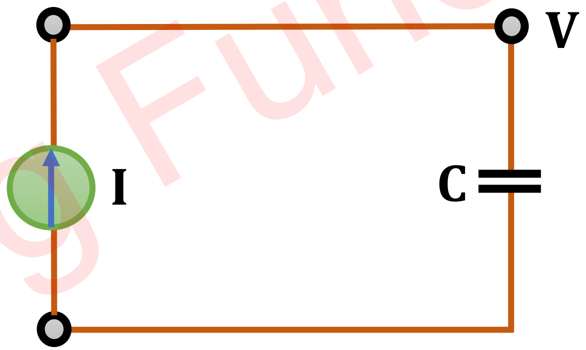
- Input = Torque τ
- Output = Angular Velocity ω



$$\Rightarrow \tau = J\alpha = J\frac{d\omega}{dt} = J\frac{d^2\theta}{dt^2}$$

❑ Electrical System

- Input = Current I
- Output = Voltage V

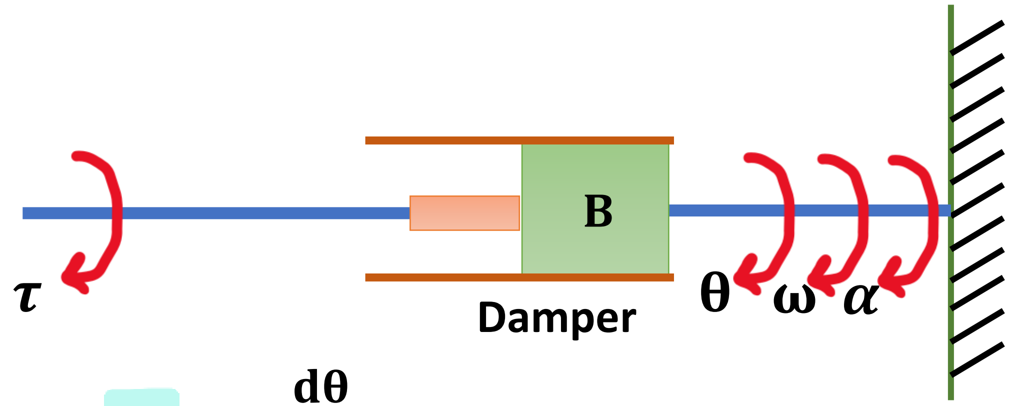


$$\Rightarrow I = C\frac{dV}{dt}$$

❑ Torque Current Analogy

Mechanical System	Electrical System
❖ Torque τ	❖ Current I
❖ Angular Velocity ω	❖ Voltage V
❖ Moment Inertia J	❖ Capacitance C

❑ Mechanical System

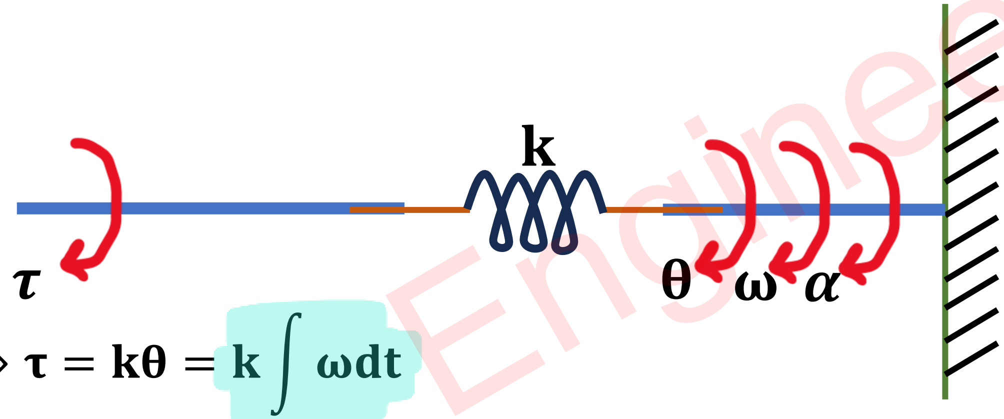


$$\Rightarrow \tau = B\omega = B\frac{d\theta}{dt}$$

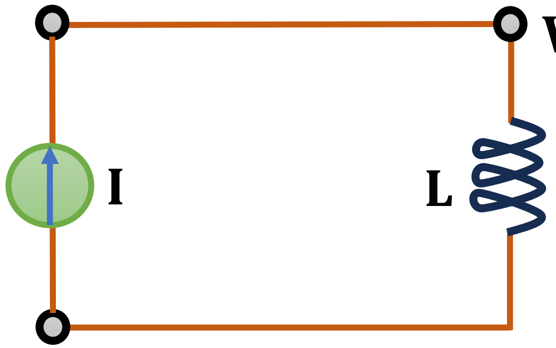
❑ Electrical System

$$\Rightarrow I = \frac{V}{R}$$

Mechanical System	Electrical System
❖ Damping Constant B	❖ Reciprocal of Resistance (1/R)



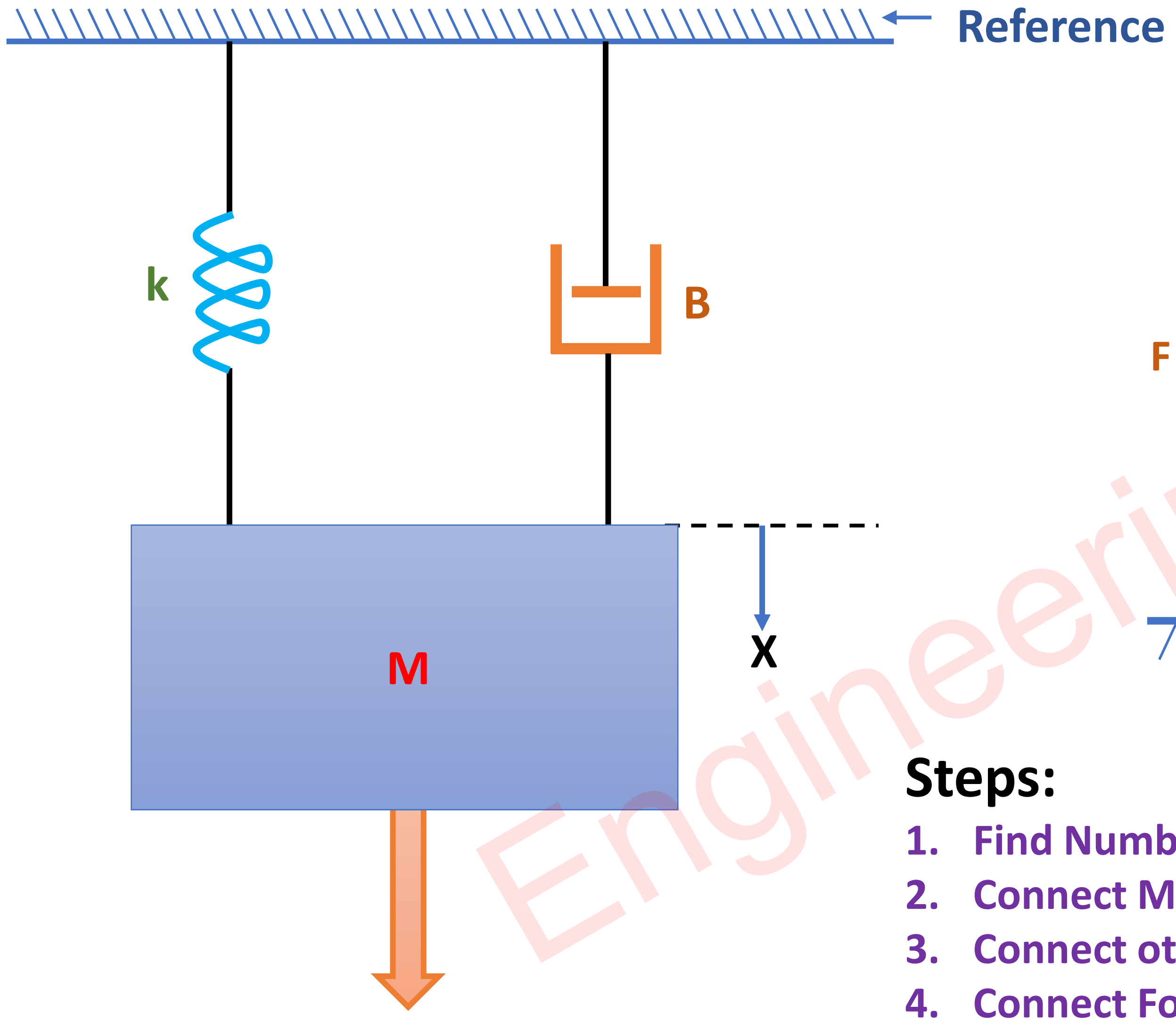
$$\Rightarrow \tau = k\theta = k\int \omega dt$$

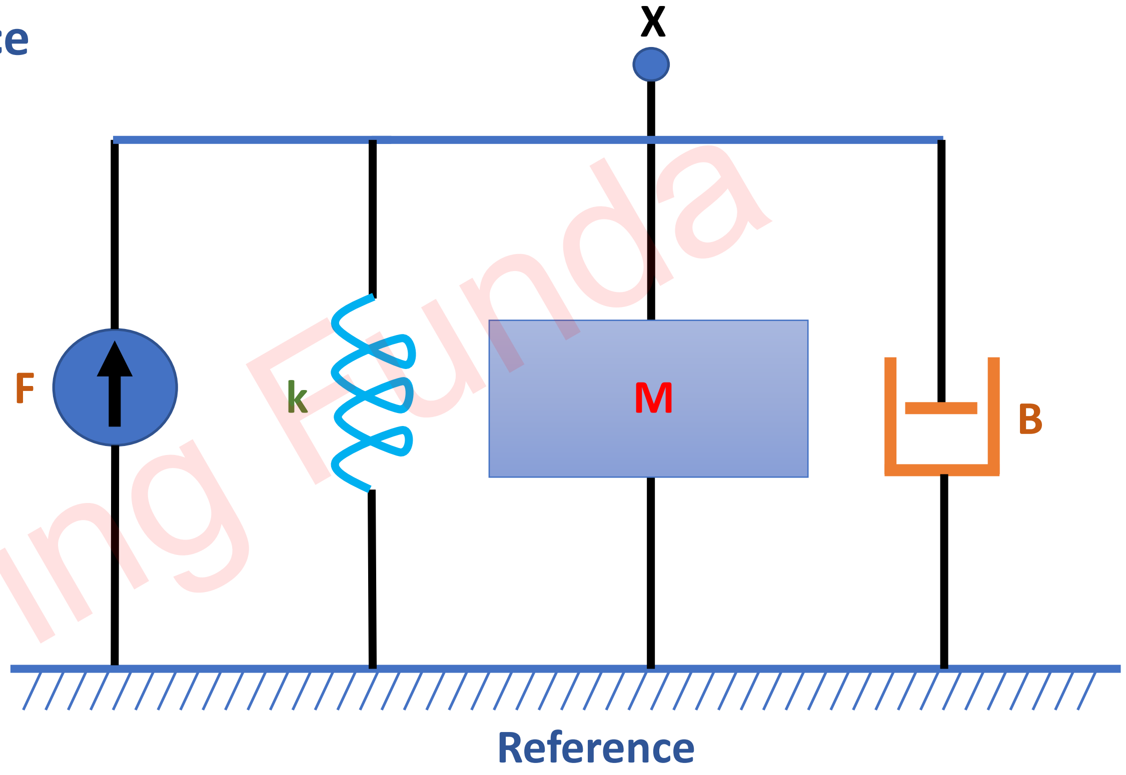


$$\Rightarrow I = \frac{1}{L}\int V dt$$

Mechanical System	Electrical System
❖ Spring Constant k	❖ Reciprocal of Inductance (1/L)

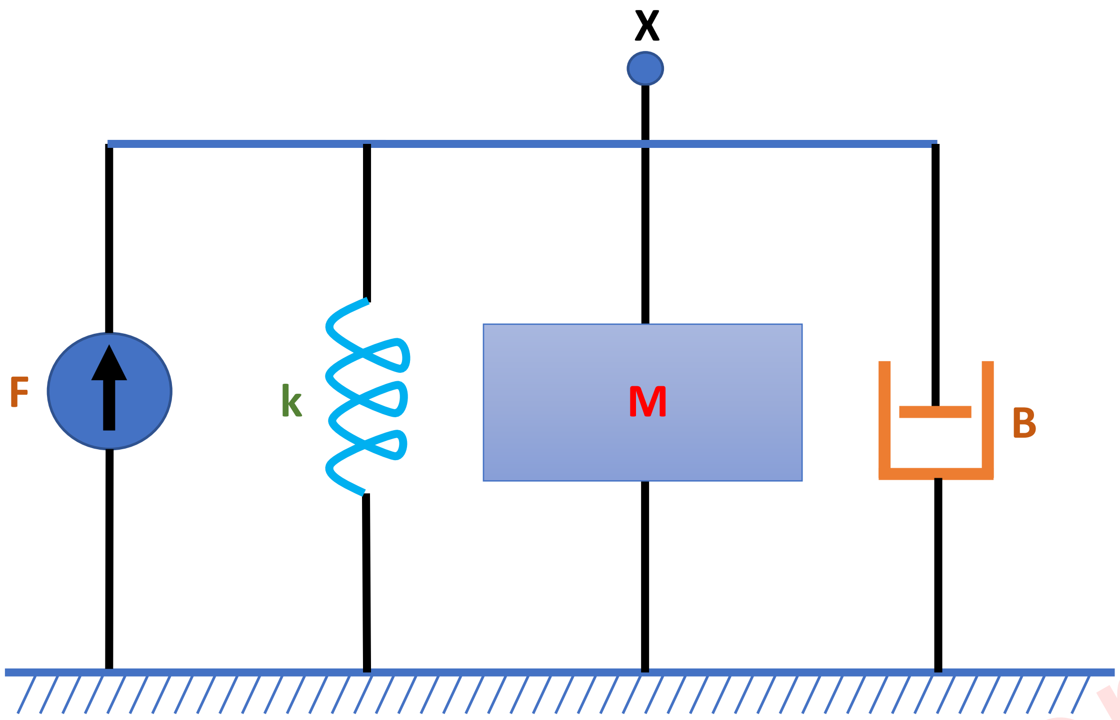
Example of Mathematical Modelling of Mechanical System





Steps:

1. Find Number of Displacement = Total Nodes = 1
2. Connect Masses from Node to Reference
3. Connect other elements as per the Model given to us
4. Connect Force towards the direction of the Node
5. Write Equations of Nodes as per Incoming Forces = Outgoing Forces



- At node x :
  - Incoming force $= F$
  - Outgoing force $= Ma + Bv + kx$
- So equation will be

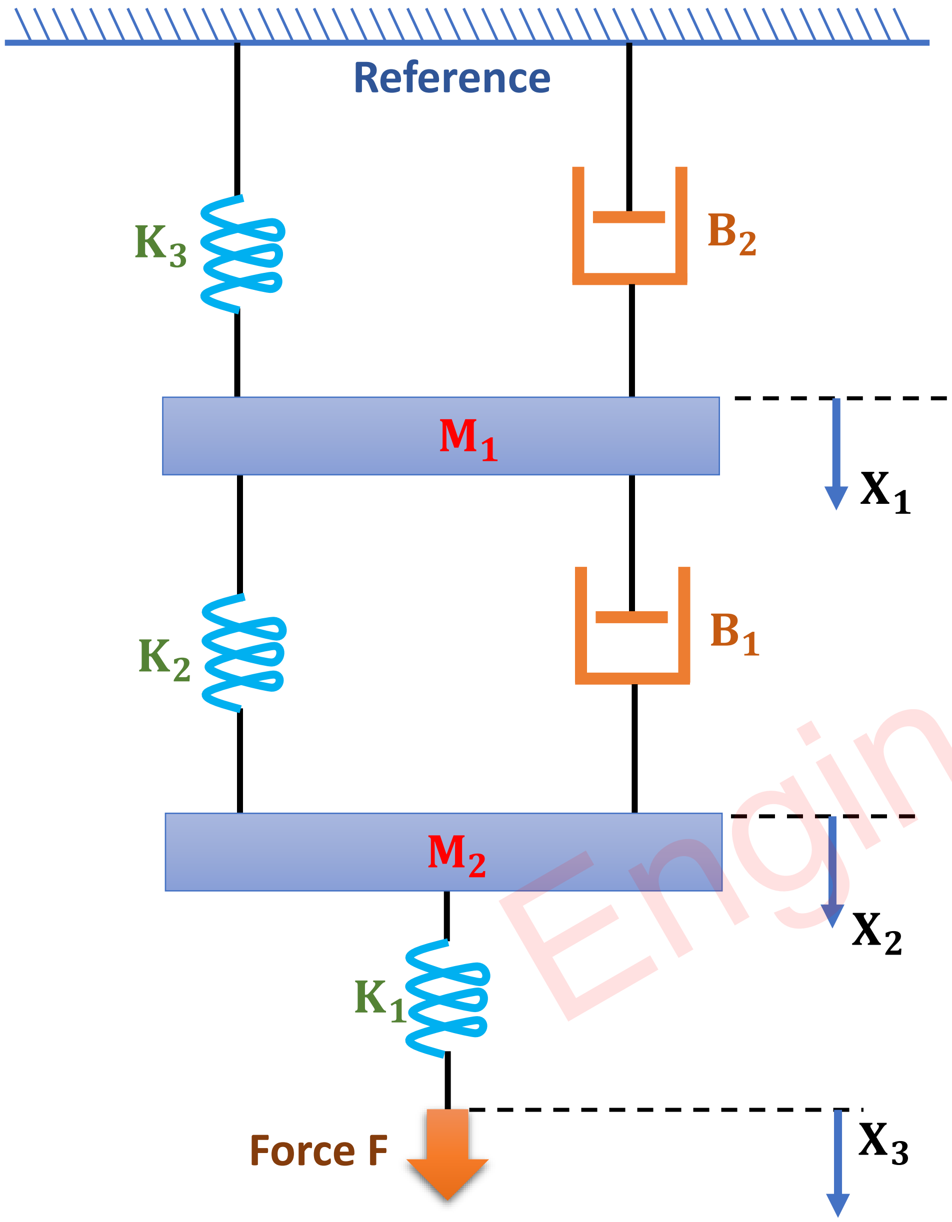
$$\therefore F = Ma + Bv + kx$$

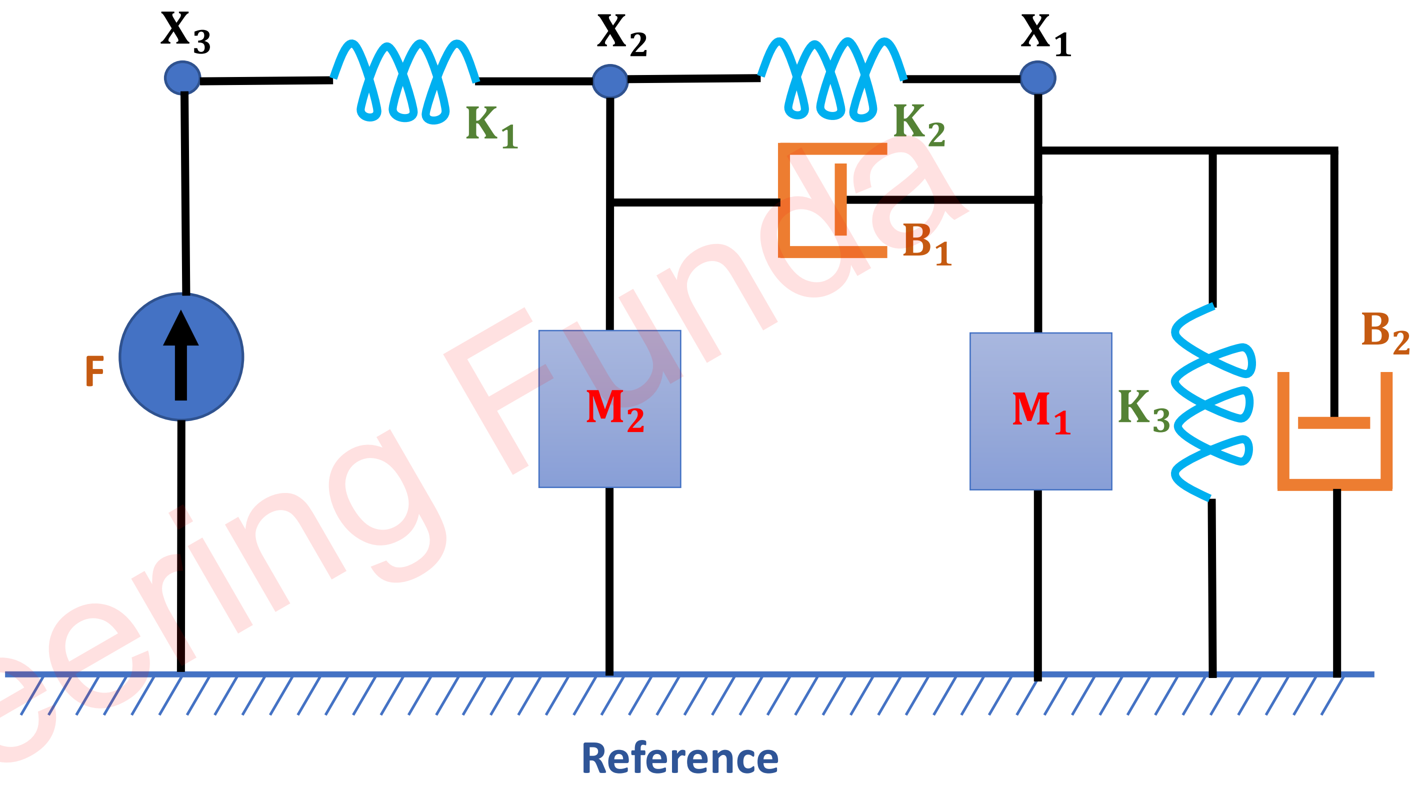
$$\therefore F = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + kx$$

- Apply Laplace transform

$$\therefore F = MS^2X(s) + BSX(s) + kX(s)$$

Example of Mathematical Modelling of Mechanical System





Steps

1. Find Number of Displacement = Total Nodes = 3
2. Connect Masses from Node to Reference
3. Connect other elements as per the Model given to us
4. Connect Force towards the direction of the Node
5. Write Equations of Nodes as per Incoming Forces = Outgoing Forces

- At node X_3:

$$\Rightarrow F = K_1(X_3 - X_2)$$

- At node X_2:

$$\Rightarrow 0 = K_1(X_2 - X_3) + K_2(X_2 - X_1) + B_1(V_2 - V_1) + M_2 a_2$$

$$\Rightarrow 0 = K_1(X_2 - X_3) + K_2(X_2 - X_1) + B_1\left(\frac{dX_2}{dt} - \frac{dX_1}{dt}\right) + M_2\frac{d^2X_2}{dt^2}$$

- At node X_1:

$$\Rightarrow 0 = K_2(X_1 - X_2) + B_1(V_1 - V_2) + M_1 a_1 + K_3X_1 + B_2V_1$$

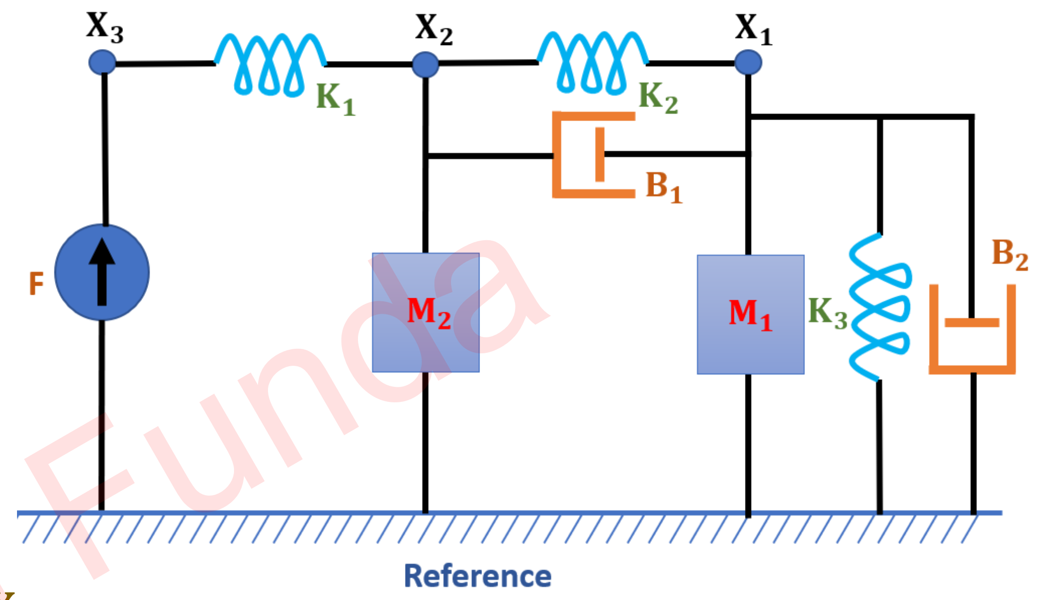
$$\Rightarrow 0 = K_2(X_1 - X_2) + B_1\left(\frac{dX_1}{dt} - \frac{dX_2}{dt}\right) + M_1\frac{d^2X_1}{dt^2} + K_3X_1 + B_2\frac{dX_1}{dt}$$

- Apply Laplace transform

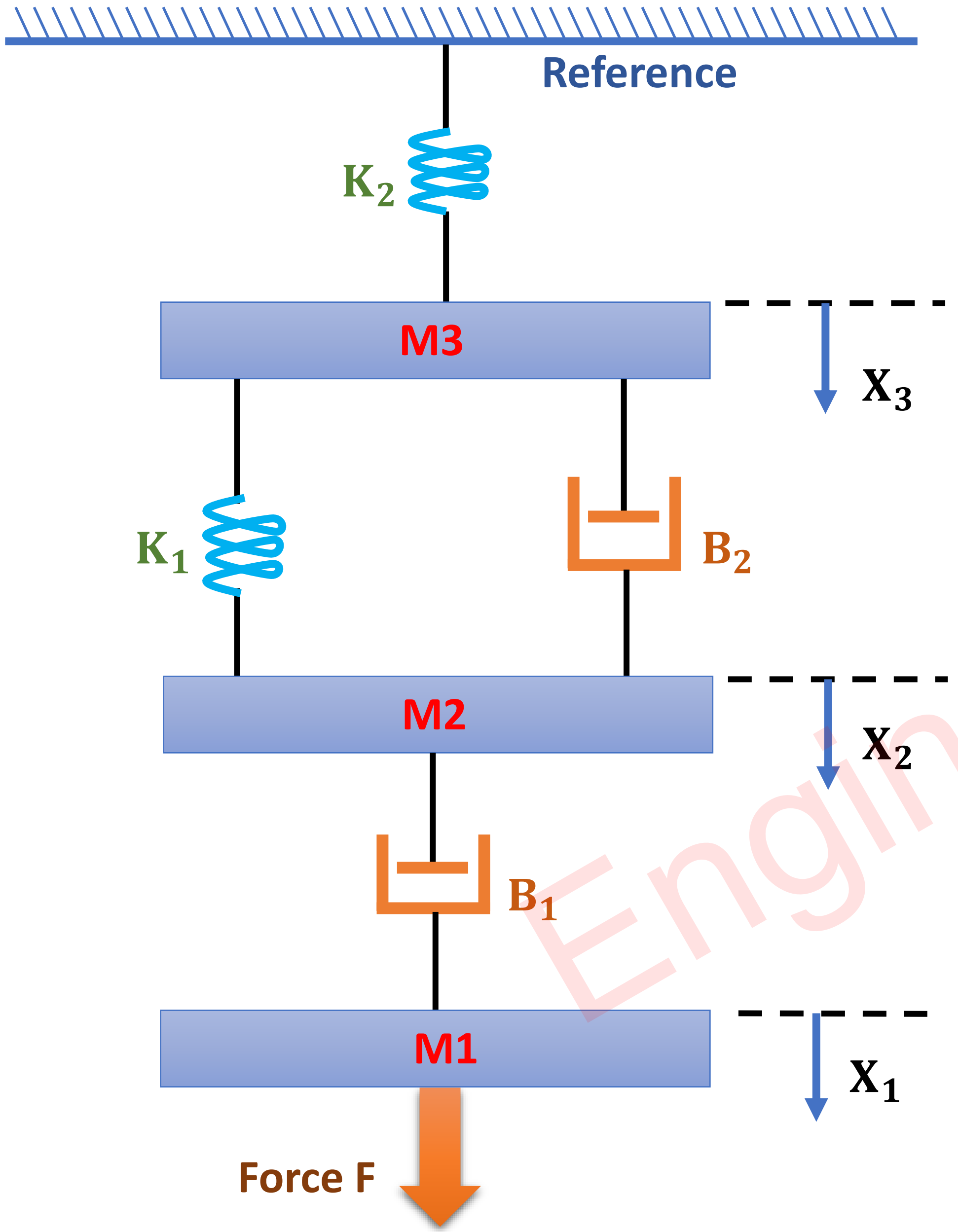
$$\Rightarrow F = K_1(X_3 - X_2)$$

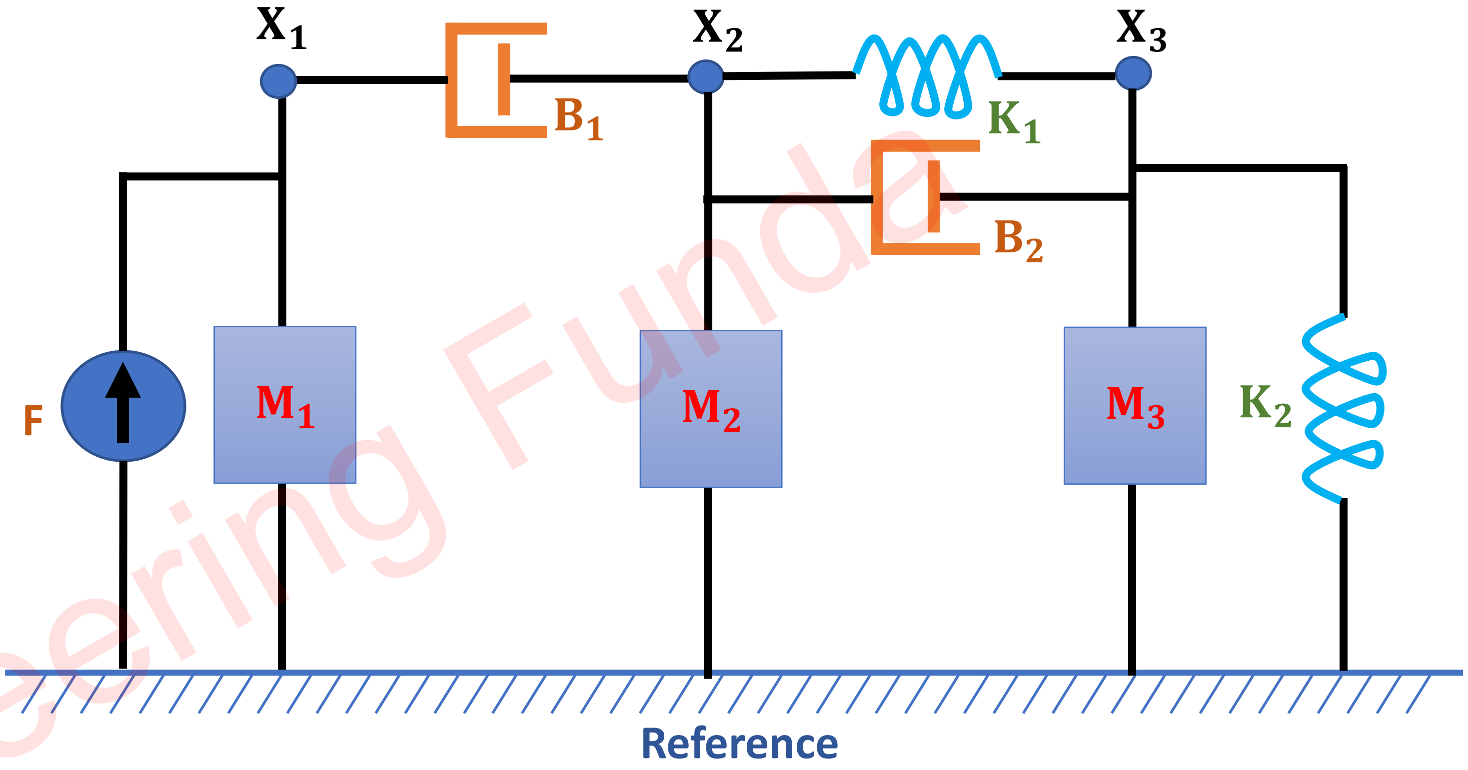
$$\Rightarrow 0 = K_1(X_2 - X_3) + K_2(X_2 - X_1) + B_1(SX_2 - SX_1) + M_2S^2X_2$$

$$\Rightarrow 0 = K_2(X_1 - X_2) + B_1(SX_1 - SX_2) + M_1S^2X_1 + K_3X_1 + B_2SX_1$$



Example of Mathematical Modelling of Mechanical System





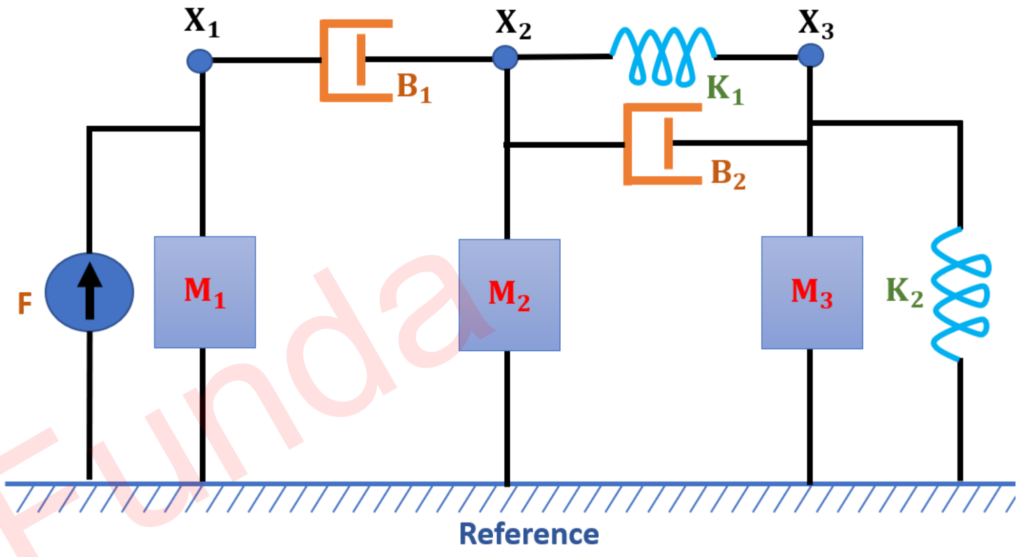
Steps

1. Find Number of Displacement = Total Nodes = 3
2. Connect Masses from Node to Reference
3. Connect other elements as per the Model given to us
4. Connect Force towards the direction of the Node
5. Write Equations of Nodes as per Incoming Forces = Outgoing Forces

- At node X_1:

$$\Rightarrow F = B_1(V_1 - V_2) + M_1 a_1$$

$$\Rightarrow F = B_1\left(\frac{dX_1}{dt} - \frac{dX_2}{dt}\right) + M_1 \frac{d^2X_1}{dt^2}$$



- At node X_2:

$$\Rightarrow 0 = K_1(X_2 - X_3) + B_2(V_2 - V_3) + B_1(V_2 - V_1) + M_2 a_2$$

$$\Rightarrow 0 = K_1(X_2 - X_3) + B_2\left(\frac{dX_2}{dt} - \frac{dX_3}{dt}\right) + B_1\left(\frac{dX_2}{dt} - \frac{dX_1}{dt}\right) + M_2 \frac{d^2X_2}{dt^2}$$

- At node X_3:

$$\Rightarrow 0 = K_1(X_3 - X_2) + B_2(V_3 - V_2) + K_2 X_3 + M_3 a_3$$

$$\Rightarrow 0 = K_1(X_3 - X_2) + B_2\left(\frac{dX_3}{dt} - \frac{dX_2}{dt}\right) + K_2 X_3 + M_3 \frac{d^2X_3}{dt^2}$$

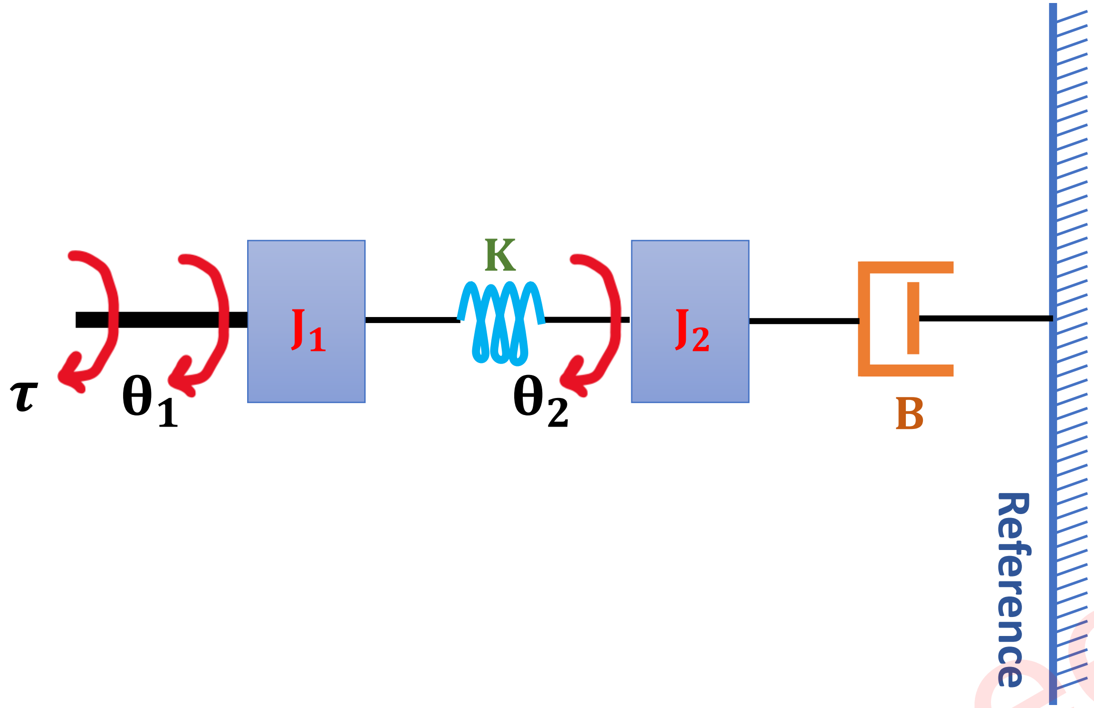
- Apply Laplace transform

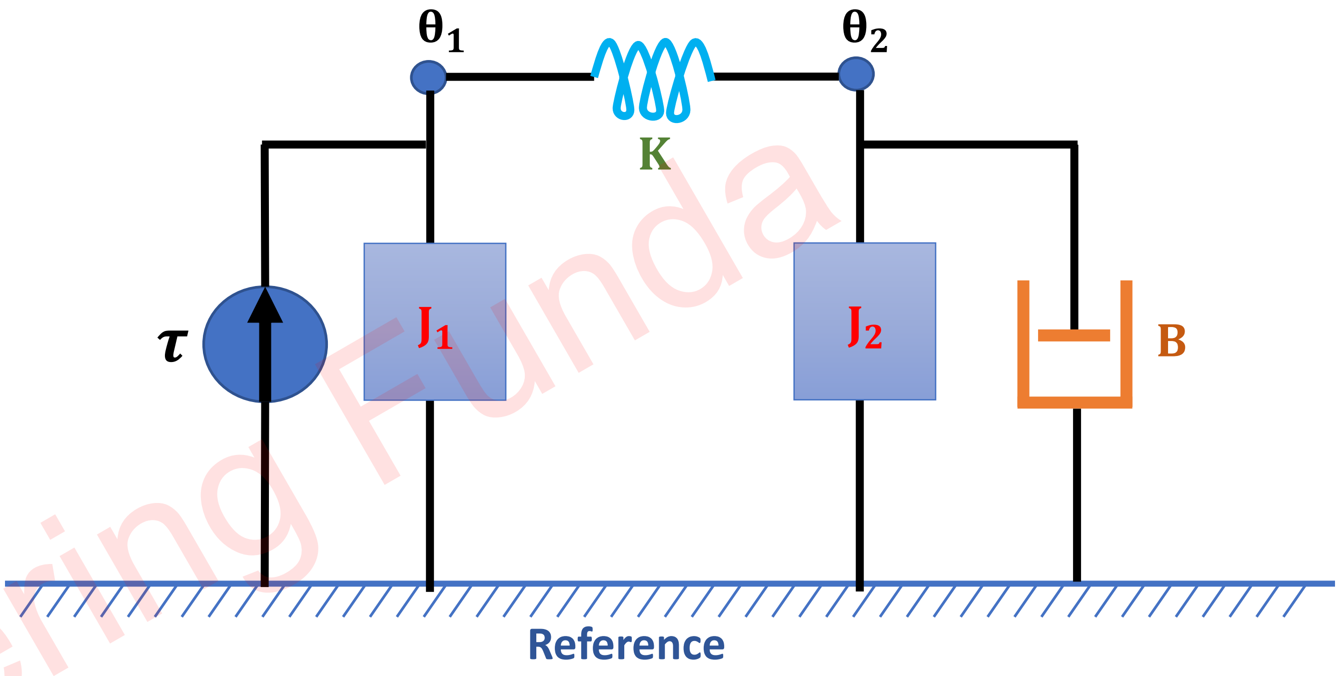
$$\Rightarrow F = B_1(SX_1 - SX_2) + M_1 S^2 X_1$$

$$\Rightarrow 0 = K_1(X_2 - X_3) + B_2(SX_2 - SX_3) + B_1(SX_2 - SX_1) + M_2 S^2 X_2$$

$$\Rightarrow 0 = K_1(X_3 - X_2) + B_2(SX_3 - SX_2) + K_2 X_3 + M_3 S^2 X_3$$

Example of Mathematical Modelling of Mechanical System





Steps

1. Find Number of Angular Displacement = Total Nodes = 2
2. Connect Moment of Inertia from Node to Reference
3. Connect other elements as per the Model given to us
4. Connect the Incoming Torque from Reference towards the direction of the Node
5. Write Equations of Nodes as per Incoming Torque = Outgoing Torque

- At node θ_1:

$$\Rightarrow \tau = K(\theta_1 - \theta_2) + J_1\alpha_1$$

$$\Rightarrow \tau = K(\theta_1 - \theta_2) + J_1\frac{d^2\theta_1}{dt^2}$$

- At node θ_2:

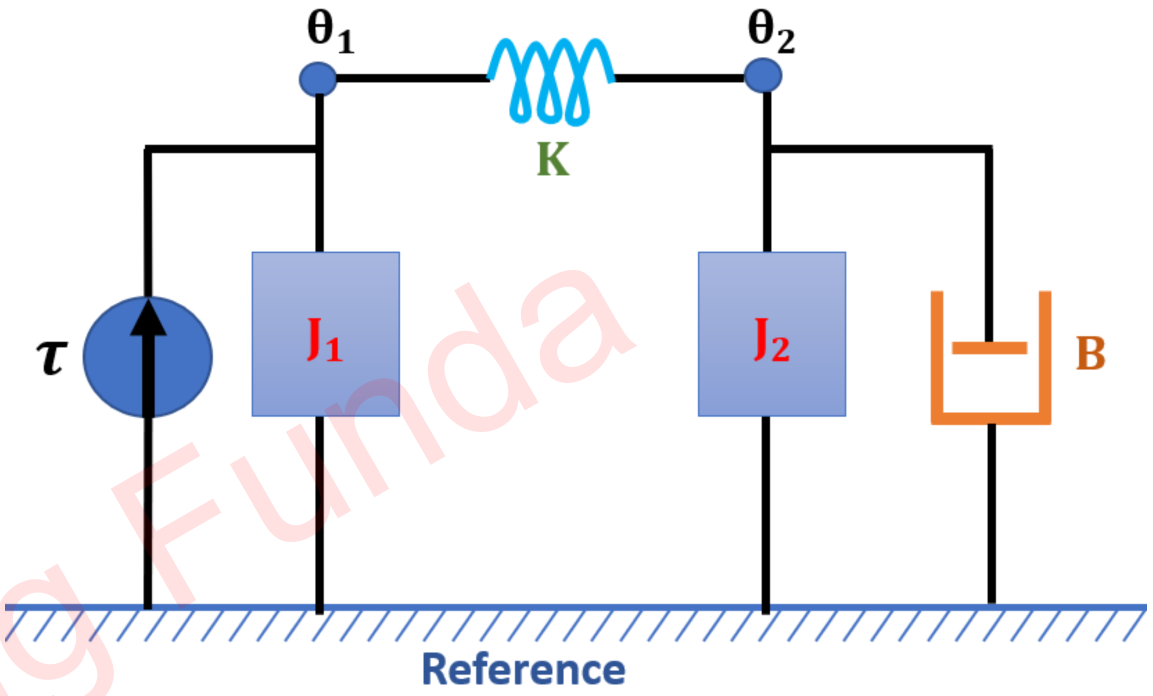
$$\Rightarrow 0 = K(\theta_2 - \theta_1) + J_2\alpha_2 + B\omega_2$$

$$\Rightarrow 0 = K(\theta_2 - \theta_1) + J_2\frac{d^2\theta_2}{dt^2} + B\frac{d\theta_2}{dt}$$

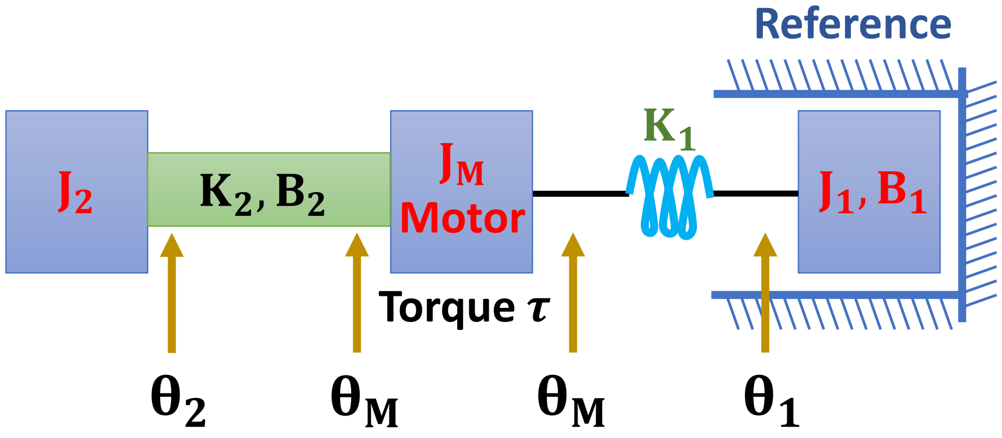
- Apply Laplace transform

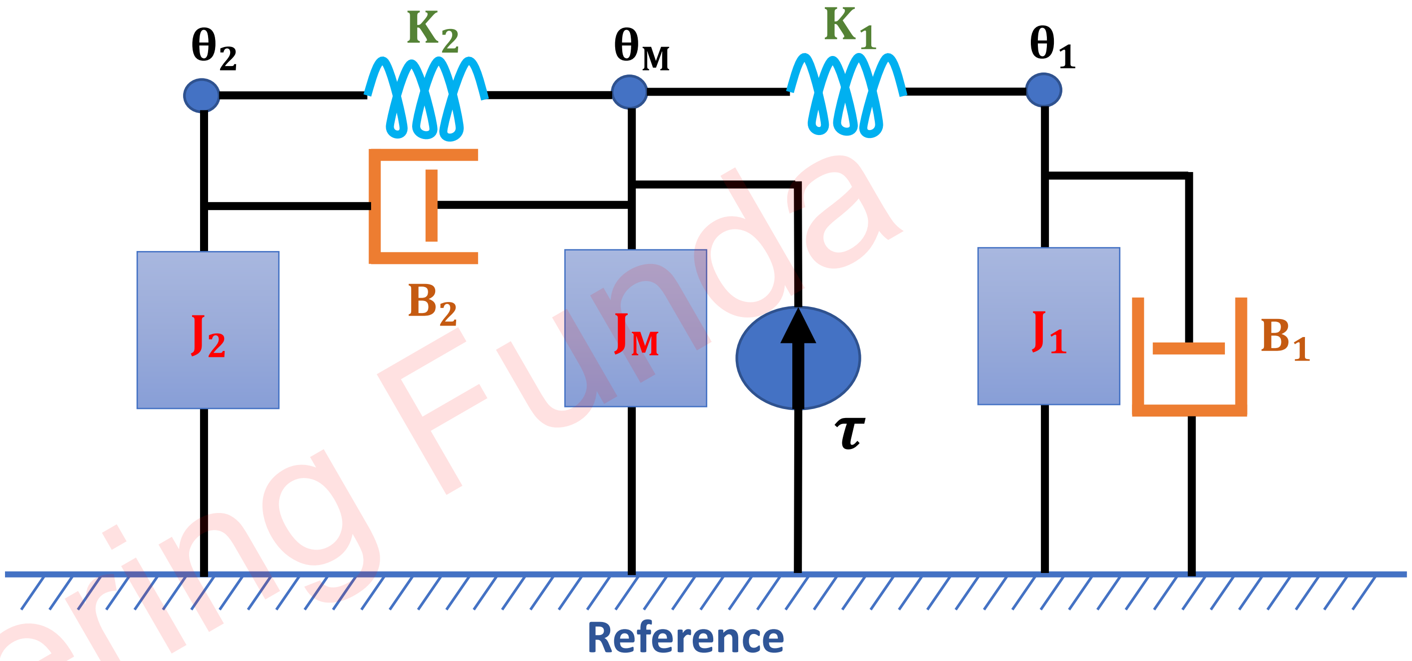
$$\Rightarrow \tau = K(\theta_1 - \theta_2) + J_1S^2\theta_1$$

$$\Rightarrow 0 = K(\theta_2 - \theta_1) + J_2S^2\theta_2 + BS\theta_2$$



Example of Mathematical Modelling of Mechanical System





Steps

1. Find Number of Angular Displacement = Total Nodes = 3
2. Connect Moment of Inertia from Node to Reference
3. Connect other elements as per the Model given to us
4. Connect the Incoming Torque from Reference towards the direction of the Node
5. Write Equations of Nodes as per Incoming Torque = Outgoing Torque

- At node θ_1:

$$\Rightarrow 0 = K_1(\theta_1 - \theta_M) + J_1\alpha_1 + B_1\omega_1$$

$$\Rightarrow 0 = K_1(\theta_1 - \theta_M) + J_1\frac{d^2\theta_1}{dt^2} + B_1\frac{d\theta_1}{dt}$$

- At node θ_2:

$$\Rightarrow 0 = J_2\alpha_2 + K_2(\theta_2 - \theta_M) + B_2(\omega_2 - \omega_M)$$

$$\Rightarrow 0 = J_2\frac{d^2\theta_2}{dt^2} + K_2(\theta_2 - \theta_M) + B_2\left(\frac{d\theta_2}{dt} - \frac{d\theta_M}{dt}\right)$$

- At node θ_M:

$$\Rightarrow \tau = J_M\alpha_M + K_2(\theta_M - \theta_2) + B_2(\omega_M - \omega_2) + K_1(\theta_M - \theta_1)$$

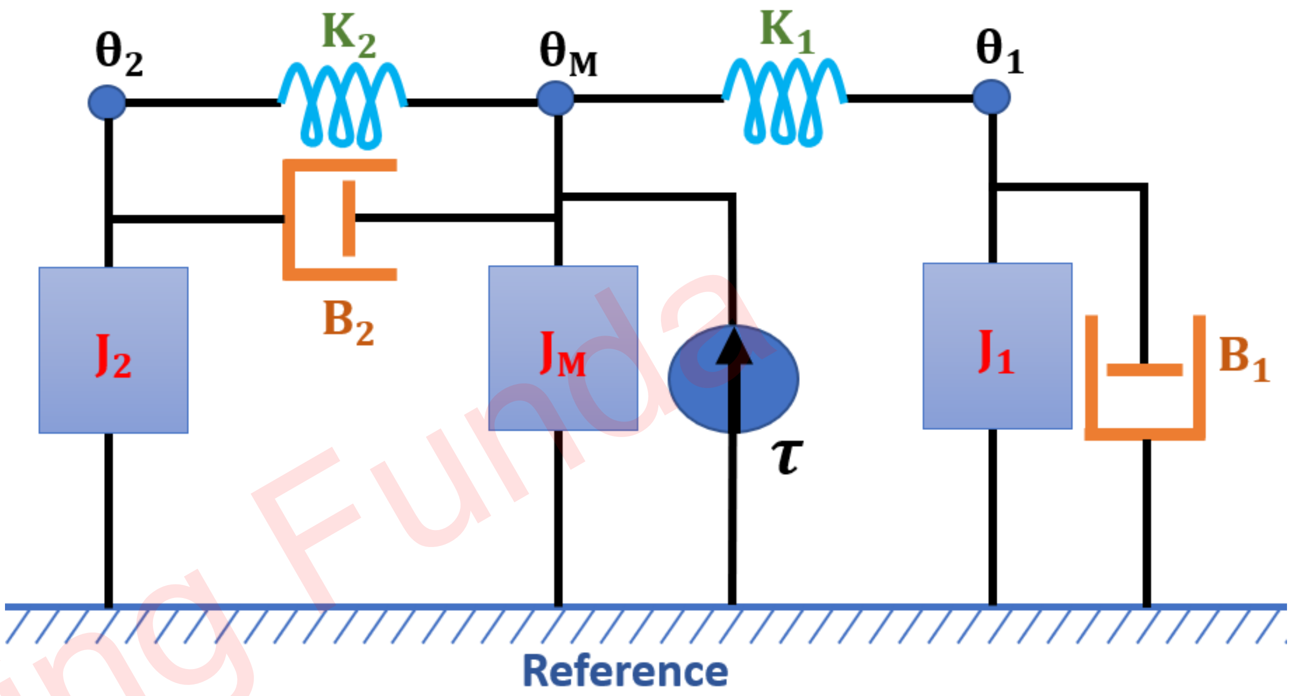
$$\Rightarrow \tau = J_M\frac{d^2\theta_M}{dt^2} + K_2(\theta_M - \theta_2) + B_2\left(\frac{d\theta_M}{dt} - \frac{d\theta_2}{dt}\right) + K_1(\theta_M - \theta_1)$$

- Apply Laplace transform

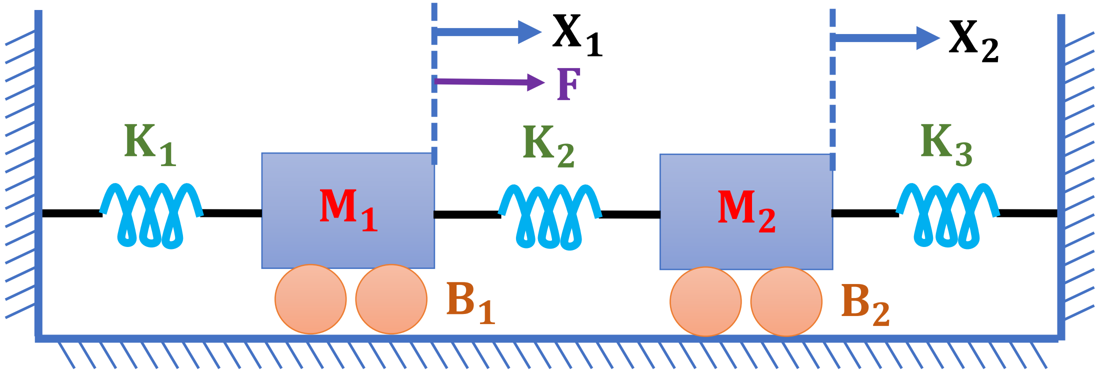
$$\Rightarrow 0 = K_1(\theta_1 - \theta_M) + J_1S^2\theta_1 + B_1S\theta_1$$

$$\Rightarrow 0 = J_2S^2\theta_2 + K_2(\theta_2 - \theta_M) + B_2(S\theta_2 - S\theta_M)$$

$$\Rightarrow \tau = J_MS^2\theta_M + K_2(\theta_M - \theta_2) + B_2(S\theta_M - S\theta_2) + K_1(\theta_M - \theta_1)$$

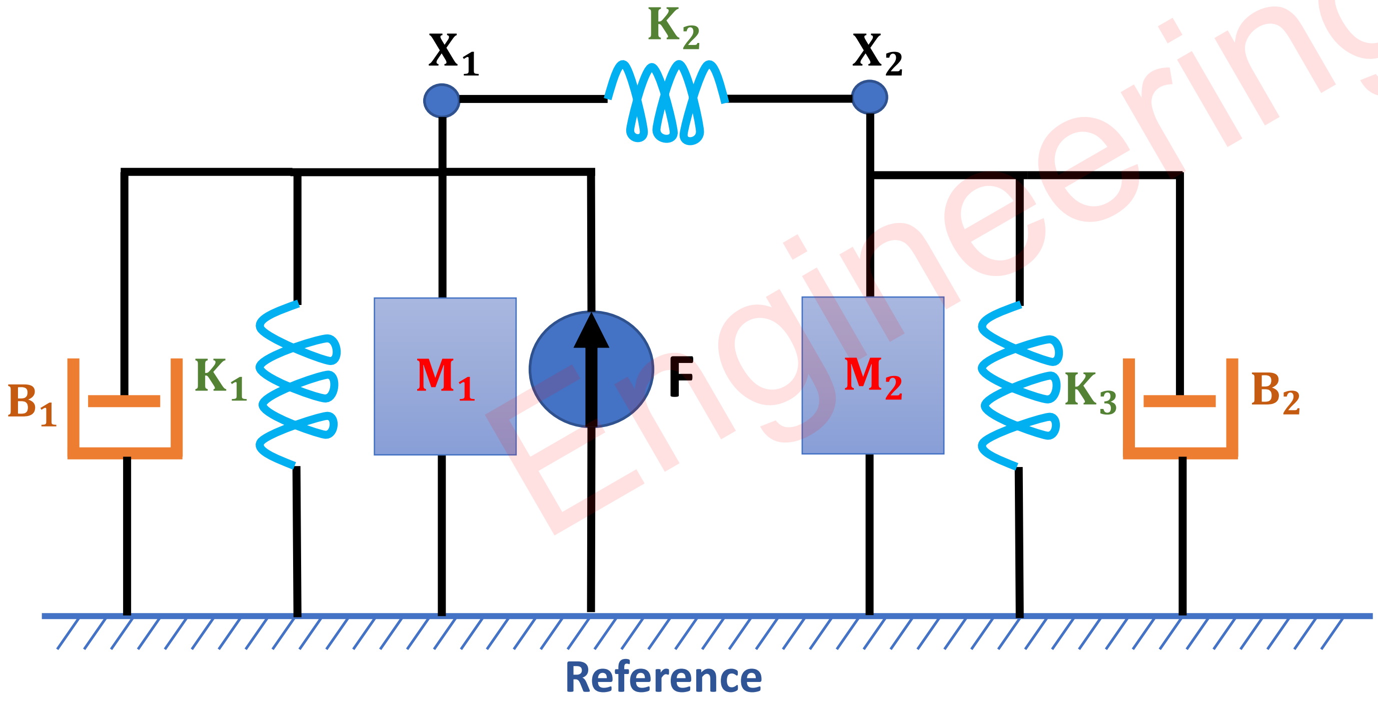


Example of Force Voltage Analogy



Steps

1. Find Number of Displacement = Total Nodes = 2
2. Connect Masses from Node to Reference
3. Connect other elements as per the Model given to us
4. Connect Force towards the direction of the Node
5. Write Equations of Nodes as per Incoming Forces = Outgoing Forces



- At node X_1:

$$\Rightarrow F = K_2(X_1 - X_2) + K_1X_1 + M_1a_1 + B_1v_1$$

$$\Rightarrow F = K_2(X_1 - X_2) + K_1X_1 + M_1\frac{d^2X_1}{dt^2} + B_1\frac{dX_1}{dt}$$

- At node X_2:

$$\Rightarrow 0 = K_2(X_2 - X_1) + K_3X_2 + M_2a_2 + B_2v_2$$

$$\Rightarrow 0 = K_2(X_2 - X_1) + K_3X_2 + M_2\frac{d^2X_2}{dt^2} + B_2\frac{dX_2}{dt}$$

- Apply Laplace transform

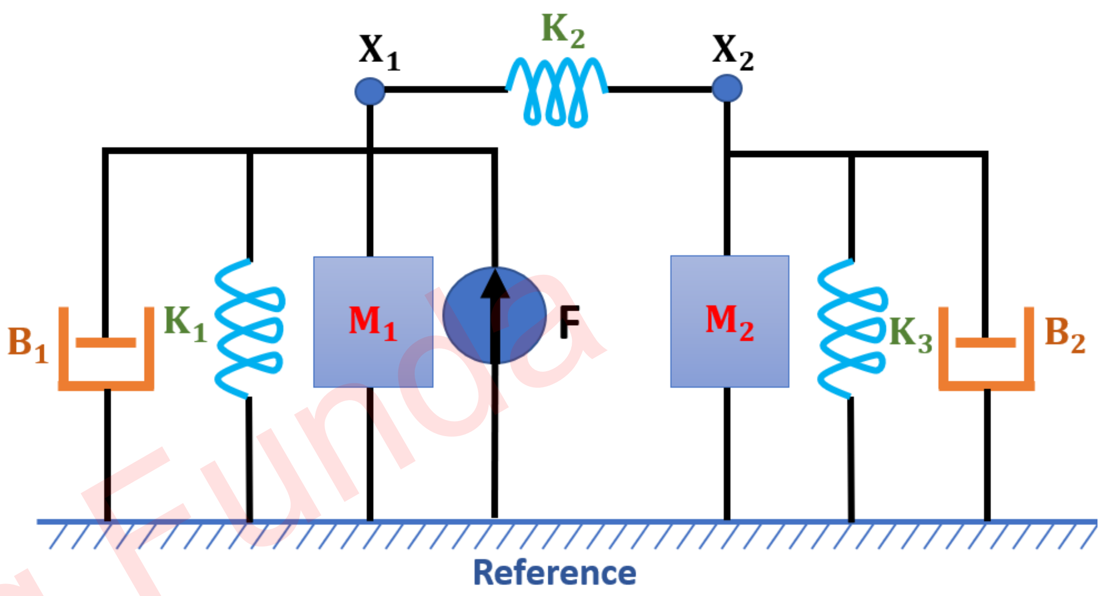
$$\Rightarrow F = K_2(X_1 - X_2) + K_1X_1 + M_1S^2X_1 + B_1SX_1$$

$$\Rightarrow 0 = K_2(X_2 - X_1) + K_3X_2 + M_2S^2X_2 + B_2SX_2$$

- Force to Voltage Analogy

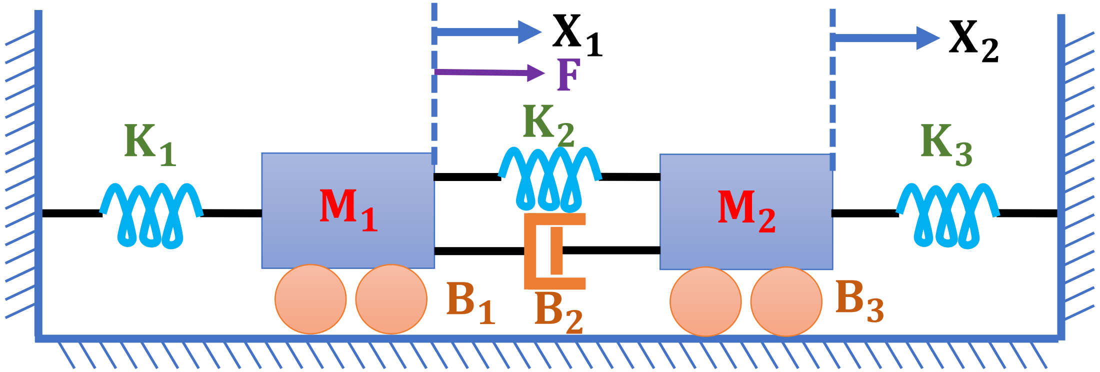
$$\Rightarrow V = \frac{1}{C_2}(Q_1 - Q_2) + \frac{1}{C_1}Q_1 + L_1S^2Q_1 + R_1SQ_1$$

$$\Rightarrow 0 = \frac{1}{C_2}(Q_2 - Q_1) + \frac{1}{C_3}Q_2 + L_2S^2Q_2 + R_2SQ_2$$



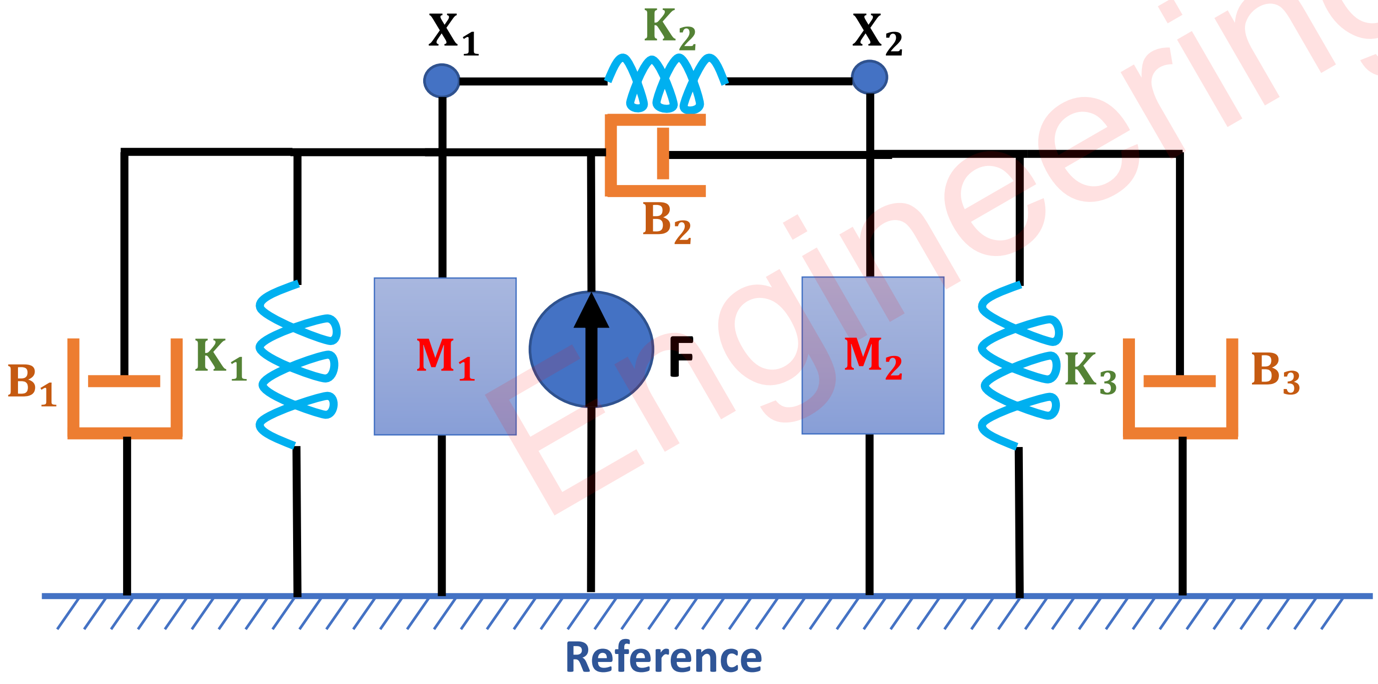
Mechanical System	Electrical System
Force F	Voltage V
Mass M	Inductance L
Damping Constant B	Resistance R
Spring Constant k	Reciprocal of Capacitance (1/C)
Distance x	Charge Q
Velocity $v = \frac{dx}{dt}$	Current $I = \frac{dQ}{dt}$

Example of Force Current Analogy



Steps

1. Find Number of Displacement = Total Nodes = 2
2. Connect Masses from Node to Reference
3. Connect other elements as per the Model given to us
4. Connect Force towards the direction of the Node
5. Write Equations of Nodes as per Incoming Forces = Outgoing Forces



- At node X_1:

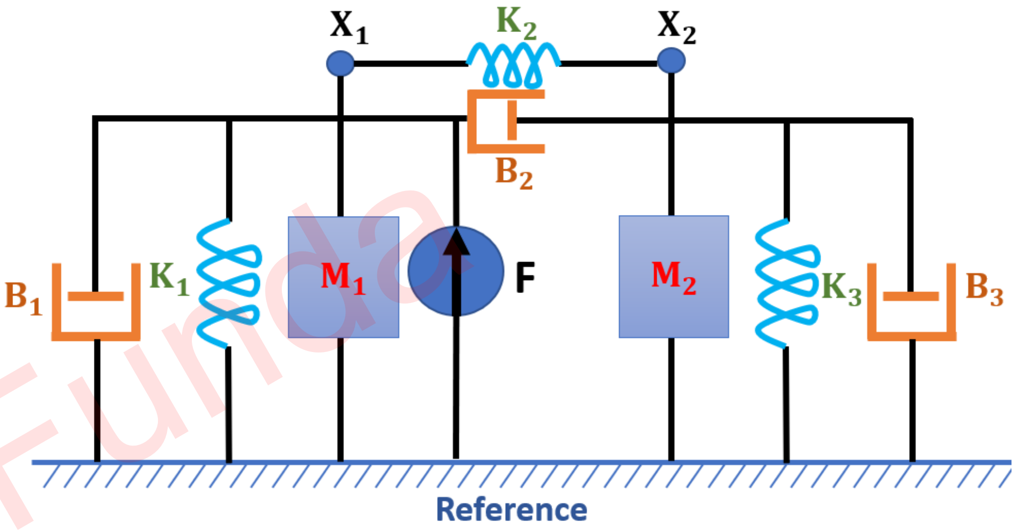
$$\Rightarrow F = K_2(X_1 - X_2) + B_2(V_1 - V_2) + K_1X_1 + M_1a_1 + B_1v_1$$

$$\Rightarrow F = K_2(X_1 - X_2) + B_2\left(\frac{dX_1}{dt} - \frac{dX_2}{dt}\right) + K_1X_1 + M_1\frac{d^2X_1}{dt^2} + B_1\frac{dX_1}{dt}$$

- At node X_2:

$$\Rightarrow 0 = K_2(X_2 - X_1) + B_2(V_2 - V_1) + K_3X_2 + M_2a_2 + B_3v_2$$

$$\Rightarrow 0 = K_2(X_2 - X_1) + B_2\left(\frac{dX_2}{dt} - \frac{dX_1}{dt}\right) + K_3X_2 + M_2\frac{d^2X_2}{dt^2} + B_3\frac{dX_2}{dt}$$



- Apply Laplace transform

$$\Rightarrow F = K_2(X_1 - X_2) + B_2(SX_1 - SX_2) + K_1X_1 + M_1S^2X_1 + B_1SX_1$$

$$\Rightarrow 0 = K_2(X_2 - X_1) + B_2(SX_2 - SX_1) + K_3X_2 + M_2S^2X_2 + B_3SX_2$$

- Force to Current Analogy

$$\Rightarrow I = \frac{1}{L_2}(\phi_1 - \phi_2) + \frac{1}{R_2}(S\phi_1 - S\phi_2) + \frac{1}{L_1}\phi_1 + C_1S^2\phi_1 + \frac{1}{R_1}S\phi_1$$

$$\Rightarrow 0 = \frac{1}{L_2}(\phi_2 - \phi_1) + \frac{1}{R_2}(S\phi_2 - S\phi_1) + \frac{1}{L_3}\phi_2 + C_2S^2\phi_2 + \frac{1}{R_3}S\phi_2$$

Mechanical System	Electrical System
❖ Force F	❖ Current I
❖ Mass M	❖ Capacitance C
❖ Damping Constant B	❖ Reciprocal of Resistance (1/R)
❖ Spring Constant k	❖ Reciprocal of Inductance (1/L)
❖ Distance x	❖ Flux ϕ
❖ Velocity $v = \frac{dx}{dt}$	❖ Voltage $V = \frac{d\phi}{dt}$